CHAPTER QUESTIONS

SKILLS INTERPRETATION (7)

1 a Draw the structural formulae for the following compounds:

i methanol **ii** propan-1-ol

b Draw the displayed formulae for these compounds:

i ethanol **ii** butan-1-ol

SKILLS CRITICAL THINKING

c Name the following compounds:

i $CH_3CH_2CH_2CH_2OH$ **ii** CH_3CH_2OH

SKILLS INTERPRETATION (8)

2 a Draw a dot-and-cross diagram showing the bonding in a molecule of ethanol.

SKILLS CRITICAL THINKING (6)

b Ethanol can be produced by the catalytic hydration of ethene. Describe two conditions used in this conversion.

c In some countries ethanol is manufactured by a different method called fermentation.

(5) **i** State the raw material used to manufacture ethanol in this way.

SKILLS REASONING (7) **ii** Explain why the following conditions are important for making ethanol by the fermentation of glucose.

- Yeast is added into the reaction mixture.
- A temperature between 30 and 40 °C is used.
- Fermentation needs to be carried out in the absence of air.

(6) **iii** Explain why some countries, such as Brazil, manufacture ethanol in this way.

SKILLS PROBLEM SOLVING (9)

d Ethanol is used as a biofuel in Brazil. Write a balanced chemical equation for the complete combustion of ethanol.

3 Glucose can be converted to ethanoic acid in the following sequence of reactions:

$$\text{glucose} \xrightarrow{\text{reaction 1}} \text{ethanol} \xrightarrow{\text{reaction 2}} \text{ethanoic acid}$$

a Write a balanced chemical equation for reaction 1 (the formula for glucose is $C_6H_{12}O_6$).

SKILLS CRITICAL THINKING

b Ethanoic acid can be made by the oxidation of ethanol in a lab. The ethanoic acid can be distilled off from the resulting mixtures and collected as a solution in water.

(5) **i** Give suitable reagents for reaction 2.

ii Describe the colour change during reaction 2.

SKILLS PROBLEM SOLVING (9) **iii** Write a balanced chemical equation for this process, using [O] to represent the oxidising agent.

(8) **c** Ethanol can be converted to ethene and another product in a single step. Complete the equation below.

$C_2H_5OH \rightarrow C_2H_4 +$ _____

END OF CHEMISTRY ONLY